មេរៀនទី០៤

ដេរីវេនៃអនុគមន៍

1. ចូរគណនាដេរីវេនៃអនុគមន៍ខាងក្រោម៖

1. $y=2019$ 2. $y=-\frac{2018}{2019}$ 3. $y=2x+3$

4. $y=-5x+4$ 5. $y=4-5x$ 6. $y=x^2+3x-5$

7. $y=2x^2-5x+7$ 8. $y=-3x^2-4x+9$ 9. $y=2x^3-3x^2+5x-4$

10. $y=x^3-3x+2$ 11. $y=x^3-5x^2+3x+1$ 12. $y=\frac{1}{3}x^3-\frac{5}{2}x^2+4x-7$

13. $y=-2x^4+3x^2-1$ 14. $y=(x+1)^2$ 15. $y=2(1-x)^2$

16. $y=(x+2)^{2019}$ 17. $y=(2x^3-3x^2+6x-1)^{15}$ 18. $y=(1-2x)^{2019}$

19. $y=(2x-4)^2-2(x-3)$ 20. $y=x+\sqrt{x}-1$ 20. $y=\sqrt{x}-2x+3$

. 21. $y=\sqrt{x-2}$ 22. $y=\sqrt{x^2-3x+1}$ 23. $y=2\sqrt{1-x^2}$

24. $y=\sqrt{x+1}+\sqrt{3x-2}$ 25. $y=x+\sqrt{1+x^2}$ 26. $y=\sqrt[4]{x-4}$

27. $y=x(x-3)$ 28. $y=(x-3)(x+1)$ 29. $y=(x^2+1)(x-1)$

30. $y=x(x-9)^2$ 31. $y=(x^2-x)(x^2-5x+6)$ 32. $y=\frac{1}{x}$

33. $y=\frac{2}{x}$ 34. $y=-\frac{3}{2x}$ 35. $y=\frac{1}{x-1}$

. 36. $y=\frac{2}{4-x}$ 37. $y=\frac{x}{x-4}$ 38. $y=\frac{x-1}{x+1}$

39. $y=\frac{x+1}{x-2}$ 40. $y=\frac{2x-3}{1-3x}$ 41. $y=\frac{2x+1}{x^2-1}$

. 42. $y=\frac{x^2+x+1}{x-1}$ 43. $y=\frac{x^2+x-1}{x^2+x+1}$ 44. $y=x+2-\frac{4}{x-2}$

45. $y=2x+1-\frac{3}{x-1}$ 46. $y=-x+5+\frac{3}{2-x}$

2. ចូរគណនាដេរីវេនៃអនុគមន៍ខាងក្រោម៖

1. $y=3\sin x$ 2. $y=2\cos x$ 3. $y=x-\sin x$

4. $y=2x-3\cos x$ 5. $y=\sin x+\cos x$ 6. $y=\sin 2x$

7. $y=\cos 3x$ 8. $y=3\cos 2x-2\sin 3x$ 9. $y=\sin^2 x$

10. $y=3\sin 4x-\cos(x-3)$ 11. $y=2\tan x-x+1$ 12. $y=3-\cot x$

13. $y=\tan^2 x$ 14 . $y=3\cot^2 x-x^2$ 15. $y=2x-x\sin 2x$

16. $y=(1-2x)\cos 3\mathrm{x}$ 17. $y=\dfrac{\sin x}{\cos x+1}$ 18. $y=\dfrac{1-\sin x}{1+\sin x}$

19. $y=\dfrac{\tan x}{x+1}$ 20. $y=3e^x$ 21. $y=2+3x-e^x$

22. $y=2e^{-x}-x$ 23. $y=e^{x-2}-3x+x^2$ 24. $y=e^{3x^2-3x+2}+e^{-2x}$

25. $y=xe^x-1$ 26. $y=(2x+1)e^{2x}$ 27. $y=\dfrac{e^x}{e^x+1}$

28. $y=\dfrac{1+e^x}{1-e^{-x}}$ 29. $y=\dfrac{e^{2x}-1}{x^2}$ 30. $y=\dfrac{e^x}{\sin x}$

31. $y=e^{-2x}(\sin x+\cos x)$ 32. $y=x+1-\dfrac{x}{e^x}$ 33. $y=(x^2-x+1)e^x$

34. $y=2\ln x-1$ 35. $y=x-\ln x+1$ 36. $y=x\ln x-1$

37. $y=(x^2+x+2)\ln x$ 38. $y=e^x\ln x$ 39. $y=\ln(x+1)$

40. $y=x-6\ln 4x$ 41. $y=\ln^2 x-\ln x$ 42. $y=\dfrac{\ln x}{x}$

43. $y=\dfrac{\ln x+1}{1-\ln x}$ 44. $y=\dfrac{\ln(x+1)}{x+1}$ 45. $y=e^{2x}\ln x$

46. $y=\ln x-\ln(x+1)$ 47. $y=\ln\left(\dfrac{x-1}{x+1}\right)$ 48. $y=2\ln\left(\dfrac{2+x}{2-x}\right)$

3. ចូរបង្ហាញសមភាពខាងក្រោម៖

1. គេមានអនុគមន៍ $f(x)=\dfrac{x^2-4x+8}{x-2}$ ។ បង្ហាញថា $\forall x\in\mathbb{R}-\{2\}\,;\,f'(x)=\dfrac{x(x-4)}{(x-2)^2}$ ។

2. មានអនុគមន៍ $f(x)=\dfrac{x^2+3x+3}{x+2}$ ។ បង្ហាញថា $\forall x\in\mathbb{R}-\{-2\}\,;\,f'(x)=\dfrac{(x+3)(x+1)}{(x+2)^2}$

3. គេមានអនុគមន៍ $f(x)=\dfrac{x^2-9}{4-x^2}$ ។ បង្ហាញថា $\forall x\in\mathbb{R}-\{-2,2\}\,;\,f'(x)=\dfrac{-10x}{\left(4-x^2\right)^2}$ ។

4. គេមានអនុគមន៍ $f(x)=\dfrac{x}{e^x}$ ។ បង្ហាញថា $\forall x\in\mathbb{R}\,;\,f'(x)=\dfrac{1-x}{e^x}$ ។

5. គេមានអនុគមន៍ $f(x)=\frac{e^x}{x^2}$ ។ បង្ហាញថា $\forall x\in\mathbb{R}-\{0\}\,;\,f'(x)=\frac{(x-2)e^x}{x^3}$ ។

6. គេមានអនុគមន៍ $f(x)=\ln\left(\frac{e^x}{e^x+1}\right)$ ។ បង្ហាញថា $\forall x\in\mathbb{R}\,;\,f'(x)=\frac{1}{1+e^x}$ ។

7. គេមានអនុគមន៍ $f(x)=\ln\left(\frac{x-3}{x+2}\right)$ ។ បង្ហាញថា $f'(x)=\frac{5}{(x-3)(x+2)}$ ។

8. គេមានអនុគមន៍ $f(x)=x+\frac{e^x}{2(e^x-2)}$ ។ បង្ហាញថា $f'(x)=\frac{(e^x-1)(e^x-4)}{(e^x-2)^2}$ ។

4. ក. កំណត់ចំនួនពិត $a\,,b\,,c$ ដើម្បីឲ្យ $f(x)=\frac{x^2+2x-4}{x-1}=ax+b+\frac{c}{x-1}\;;\;x\neq 1$ ។

ខ. គណនាដេរីវេ $f'(x)\,;\,f''(x)$ និង $f'''(x)$ ។

5. គេឲ្យអនុគមន៍ $f(x)=\frac{2x-3x^2+4}{x+2}\;,\;x\neq -2$ ។ រកចំនួនពិត $A\,,B\,,C$ ដើម្បីឲ្យ $f(x)=Ax+B+\frac{C}{x+2}$ ។ រួចគណនាដេរីវេ $f'(x)\,;\,f''(x)$ និង $f'''(x)$ ។

6. គេឲ្យអនុគមន៍ $g(x)=\frac{2x-4}{x^2-1}\;,\;x\neq\pm 1$ ។

ក. កំណត់ចំនួនពិត $a\,;\,b$ ដែល $g(x)=\frac{a}{x-1}+\frac{b}{x+1}$ ។

ខ. ចូរគណនាដេរីវេ $g'(x)\,;\,g''(x)$ និង $g'''(x)$ ។

7. គេឲ្យអនុគមន៍ $h(x)=\frac{5x+4}{(x+2)(x-1)}$ កំណត់ចំពោះ $x\in\mathbb{R}-\{-2,1\}$ ។

ក. កំណត់ចំនួនពិត $A\,;\,B$ ដើម្បីឲ្យបាន $h(x)=\frac{A}{x+2}+\frac{B}{x-1}$ ។

ខ. គណនា $h'(x)\,;\,h''(x)\,;\,h'''(x)$ និង $h^{(4)}(x)$ ។

8. គេឲ្យអនុគមន៍ $f(x)=\frac{5x-1}{1-x-2x^2}$ កំណត់ចំពោះ $x\in\mathbb{R}-\left\{1,\frac{1}{2}\right\}$ ។

ក. កំណត់ចំនួនពិត $m\,;\,n$ ដើម្បីឲ្យបាន $f(x)=\frac{m}{x+1}+\frac{n}{2x-1}$ ។

ខ. គណនា $f'(x)\,;\,f''(x)\,;\,f'''(x)$ និង $f^{(4)}(x)$ ។